PHY201: Waves & Optics

Problem Set 3

August 30, 2025

1. An object of mass $m = 0.20$ kg is hung from a spring of spring constant $k = 80\ \mathrm{N\,m^{-1}}$. The body is subject to a resistive force proportional to its velocity, $F_\mathrm{d} = -b\,\dot{x}$, with $b = 4\ \mathrm{N\,s\,m^{-1}}$.

   (a) Set up the differential equation of motion for the *free* (undriven) oscillations of the system, and determine the period of these damped oscillations.

   (b) Now suppose the object is subjected to a sinusoidal driving force $F(t) = F_0 \cos(\omega t)$. In the steady state, what is the amplitude $A(\Omega)$ of the forced oscillations? Given $F_0 = 2N$ and $\omega = 30\ \mathrm{sec}^{-1}$.

2. A block of mass m is connected to a spring, the other end of which is fixed. There is also a viscous damping mechanism. The following observations have been made on this system:

   (1) If the block is pushed horizontally with a force equal to mg, the static compression of the spring is equal to h.

   (2) The viscous resistive force is equal to mg if the block moves with a certain known speed u.

   (a) For this complete system (including both spring and damper) write the differential equation governing horizontal oscillations of the mass in terms of m, g, h, and u.

   (b) What is the angular frequency of the damped oscillations if $u = 3\sqrt{gh}$?

   (c) After what time, expressed as a multiple of $\sqrt{h/g}$, is the energy down by a factor $1/e$?

   (d) What is the Q of this oscillator?

   (e) This oscillator, initially in its rest position, is suddenly set into motion at $t = 0$ by a bullet of negligible mass but non-negligible momentum traveling in the positive x direction. Find the value of the phase angle δ in the equation

   $$x = Ae^{-\gamma t/2} \cos(\omega t - \delta)$$

   that describes the subsequent motion, and sketch x versus t for the first few cycles.

   (f) If the oscillator is driven with a force $mg \cos \omega t$, where $\omega = \sqrt{2g/h}$, what is the amplitude of the steady-state response?

3. Consider a system with a damping force undergoing forced oscillations at an angular frequency ω.

   (a) What is the instantaneous kinetic energy of the system?

   (b) What is the instantaneous potential energy of the system?

   (c) What is the ratio of the average kinetic energy to the average potential energy? Express the answer in terms of the ratio ω/ω_0.

   (d) For what value(s) of ω are the average kinetic energy and the average potential energy equal? What is the total energy of the system under these conditions?

   (e) How does the total energy of the system vary with time for an arbitrary value of ω? For what value(s) of ω is the total energy constant in time?